

Exercise Series 1 (solutions)

Ex 1. $y_t = a \cos(\omega t + \phi)$, $t=1, \dots, n$
 $= a \cos(\omega t) \cos \phi - a \sin(\omega t) \sin \phi$

(a) The model is not linear due to non-linear terms in ω & ϕ .

(b) If ω is fixed / known, then $\cos(\omega t)$ & $\sin(\omega t)$ can be treated as covariates. The model is still not linear due to non-linear terms $\cos \phi$ & $\sin \phi$ in ϕ .

If, however, ω is known and we are allowed to reparameterize, then

$$y_t = a \cos \phi \cos(\omega t) - a \sin \phi \sin(\omega t)$$
$$= b_1 x_1 + b_2 x_2, \text{ where } b_1 = a \cos \phi, b_2 = a \sin \phi$$

unknown parameters

$$x_1 = \cos(\omega t), x_2 = \sin(\omega t)$$

known covariates

This is a linear model with parameters b_1 & b_2 .

For part (c), you need to make the plots.

Ex 2 (a) linear model with parameters $\theta_0, \theta_1, \theta_2$ and covariates $1, \frac{1}{x}, \frac{1}{x^2}$.

(b) NOT linear. Cannot be reparameterized to become linear.

(c) NOT linear. Can be made linear by considering

$$\beta = \frac{\theta_0}{\theta_1}, z = \frac{1}{x}. \text{ Then the model becomes } y = \beta z + \varepsilon$$

(d) NOT linear. Cannot be made linear.

(e) NOT linear. Cannot be made linear.

(f) NOT linear. Cannot be made linear.

(g) NOT linear, But

$$\underbrace{\log y}_{y'} = \underbrace{\log \theta_1}_{\theta_0'} + \underbrace{\theta_2 \log x_1}_{\theta_1' x_1'} + \underbrace{\theta_3 \log (\cos x_2)}_{\theta_2' x_2'} + \underbrace{\log \varepsilon}_{\varepsilon'}$$

(h) NOT linear. Cannot be made linear.

Ex 3. $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \varepsilon$, $E(\varepsilon) = 0$.

$x_1 = \mathbb{1}_{\{\text{class } \bar{X}_{II}\}}$, $x_2 = \mathbb{1}_{\{\text{bachelor's}\}}$, $x_3 = \mathbb{1}_{\{\text{master's}\}}$

(a) Vector matrix form: $\underline{y} = X \underline{\theta} + \underline{\varepsilon}$

$$X_{8 \times 4} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \underline{\theta} = \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix}, \underline{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_8 \end{pmatrix}$$

$$\underline{y} = [50000, 35000, 80000, 90000, 95000, 25000, 30000, 20000]^T.$$

(b) $E(y) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$

expected salary =
$$\begin{cases} \theta_0 & \text{if } \text{Edu} < \text{class } \bar{X}_{II} \\ \theta_0 + \theta_1 & \text{if } \text{class } \bar{X}_{II} \leq \text{Edu} < \text{Bachelor's} \\ \theta_0 + \theta_1 + \theta_2 & \text{if } \text{Bachelor's} \leq \text{Edu} < \text{Master's} \\ \theta_0 + \theta_1 + \theta_2 + \theta_3 & \text{if } \text{Edu} \geq \text{Master's} \end{cases}$$

θ_0 is the expected base salary with less than class $\bar{x}_H$ education.

θ_1 is the expected increase in salary due to class $\bar{x}_H$ education.

θ_2 _____ college education.

θ_3 _____ university education

In each stage, the expected increase is w.r.t. the previous stage.

(c) Does salary depend on the level of education

$$\Leftrightarrow \theta_0 = \theta_0 + \theta_1 = \theta_0 + \theta_1 + \theta_2 = \theta_0 + \theta_1 + \theta_2 + \theta_3 ?$$

$$\Leftrightarrow \theta_1 = \theta_2 = \theta_3 = 0$$

So, we formulate

$$H_0: \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \underline{0} \quad \text{vs.} \quad H_a: \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} \neq \underline{0}$$

$$(d) \quad \underline{y} = \gamma_1 x_1 + \gamma_2 x_2 + \gamma_3 x_3 + \varepsilon$$

$\Uparrow$

$$\underline{y} = \underline{Z} \underline{\gamma} + \underline{\varepsilon}, \quad \underline{y} \text{ and } \underline{\varepsilon} \text{ are as before}$$

$$\underline{Z} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \underline{\gamma} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{pmatrix}$$

columns are the 2nd, 3rd and 4th columns of X

(e) In the new model

$$E(\underline{y}) = \begin{cases} 0 & \text{if } \text{Edu} < \text{class } \underline{XII} \\ \gamma_1 & \text{if } \text{class } \underline{XII} \leq \text{Edu} < \text{Bachelor's} \\ \gamma_1 + \gamma_2 & \text{if } \text{Bachelor's} \leq \text{Edu} < \text{Master's} \\ \gamma_1 + \gamma_2 + \gamma_3 & \text{if } \text{Edu} > \text{Master's} \end{cases}$$

γ_1 is the expected salary for a person who completed class XII but didn't complete bachelor's

γ_2 is the expected increase in salary if completed bachelor's but not masters.

γ_3 is the expected increase in salary (compared to the previous stage) if completed master's

(f) Connection between parameters:

$$\left. \begin{aligned} \gamma_1 &= \theta_0 + \theta_1 \\ \gamma_2 &= \theta_0 + \theta_1 + \theta_2 \\ \gamma_3 &= \theta_0 + \theta_1 + \theta_2 + \theta_3 \end{aligned} \right\} \begin{aligned} \theta_2 &= \gamma_2 - \gamma_1 \\ \theta_3 &= \gamma_3 - \gamma_2 \\ \theta_0 + \theta_1 &= \gamma_1 \end{aligned}$$

We can get to $(\gamma_1, \gamma_2, \gamma_3)$ from $(\theta_0, \theta_1, \theta_2, \theta_3)$ but only part of $(\theta_0, \theta_1, \theta_2, \theta_3)$ can be obtained from $(\gamma_1, \gamma_2, \gamma_3)$.

(g). For $y = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \varepsilon$.

$\delta_0 \leftarrow$ expected base salary ($\text{Edu} < \underline{XII}$)

$\delta_1 \leftarrow$ increase in base salary due to class $\underline{XII}$ but less than bachelors education

$\delta_2 \leftarrow$ increase in salary in the previous stage due to Bachelors but less than masters education.

$$\left. \begin{array}{l} \delta_0 = \theta_0 \\ \delta_0 + \delta_1 = \theta_0 + \theta_1 \\ \delta_0 + \delta_1 + \delta_2 = \theta_0 + \theta_1 + \theta_2 \end{array} \right\} \begin{array}{l} \theta_0 = \delta_0 \\ \theta_1 = \delta_1 \\ \theta_2 = \delta_2 \end{array} \quad \left. \vphantom{\begin{array}{l} \delta_0 = \theta_0 \\ \delta_0 + \delta_1 = \theta_0 + \theta_1 \\ \delta_0 + \delta_1 + \delta_2 = \theta_0 + \theta_1 + \theta_2 \end{array}} \right\} \begin{array}{l} \text{cannot recover} \\ \theta_3. \end{array}$$

For $y = \tau_0 + \tau_2 x_2 + \tau_3 x_3 + \varepsilon$

$$E(\text{Salary}) = \begin{cases} \tau_0 & \text{if } \text{Edu} < \text{College} \\ \tau_0 + \tau_2 & \text{if } \text{College} \leq \text{Edu} < \text{University} \\ \tau_0 + \tau_2 + \tau_3 & \text{if } \text{Edu} \geq \text{University}. \end{cases}$$

$$\left. \begin{array}{l} \tau_0 + \tau_2 = \theta_0 + \theta_1 + \theta_2 \\ \tau_0 + \tau_2 + \tau_3 = \theta_0 + \theta_1 + \theta_2 + \theta_3 \end{array} \right\}$$

Ex 5 $E(y_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = (1 \ x_{i1} \dots x_{ip}) \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}$

$$= \underline{x}_i^T \underline{\beta}$$

$$\therefore E(\underline{y}) = \begin{pmatrix} \underline{x}_1^T \\ \vdots \\ \underline{x}_n^T \end{pmatrix} \underline{\beta} = \underline{X} \underline{\beta} = f(\underline{\beta}) \text{ (say)}$$

Now, $f(c\underline{\beta}) = \underline{X}(c\underline{\beta}) = c\underline{X}\underline{\beta} = c f(\underline{\beta})$

$f(\underline{\beta}_1 + \underline{\beta}_2) = \underline{X}(\underline{\beta}_1 + \underline{\beta}_2) = \underline{X}\underline{\beta}_1 + \underline{X}\underline{\beta}_2 = f(\underline{\beta}_1) + f(\underline{\beta}_2)$

So, $E(y)$ is a linear function of $\underline{\beta}$.